Wenzhuo Zhang

(608) 867-7875 | zhangwenzhuo85@gmail.com | GitHub: MarioZhang420 | LinkedIn: Wenzhuo Zhang

EDUCATION

University of California, Berkeley — Berkeley, CA — Sep. 2025 – Dec. 2025
Visiting Student, Computer Science — AI, Operating Systems, Probability, Machine Learning

University of Wisconsin-Madison — Madison, WI — Sep. 2022 – Dec. 2026
Bachelor of Science in Computer Science — OOP, Algorithms, DBMS, Big Data Systems, UI, Machine Organization

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, C/C++, C#, SQL, TypeScript, JavaScript, Scala, HTML/CSS
Frameworks & APIs: Spring Boot, React, Node.js, Express.js, Vue.js, Go, RESTful APIs, LangChain, Gin
Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Elasticsearch, SQLite, Firebase
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Airflow
ML & AI: PyTorch, OpenCV, RAG, MLflow, Spark, OpenAI API, LangChain, Jetpack Compose
Tools & Process: Git, JWT, Agile/Scrum, Ant Design, ZooKeeper, Linux

WORK EXPERIENCE

The Trade Desk — Jun. 2026 – Aug. 2026
Software Engineer Intern — ***Scala, Python, Spark, Airflow, MLflow***

- Constructed a real-time dataset backfill platform with **Scala, Python, Spark, and Airflow**, automating workflow generation from DAG metadata.
- Shipped a DAG parsing framework converting **Airflow** workflows into JSON configs, enabling self-service dataset backfill across experiments.
- Mapped upstream and downstream ETL dependency lineage automatically, eliminating manual chain configuration across production pipelines.
- Orchestrated ETL, model training, AUC evaluation, and **MLflow** registration into a reliable end-to-end backfill system.

Tencent America LLC — Palo Alto, CA — Jan. 2026 – Apr. 2026
IT Application Developer Intern — ***Python, RAG, PostgreSQL, Redis, Jenkins, Docker, CI/CD***

- Launched internal AI apps with **RAG** and multi-agent workflows, processing 10K+ documents through embedding pipelines and hybrid search.
- Assembled a GraphRAG system integrating CMDB dependency topology and knowledge graph models for intelligent incident root-cause analysis.
- Wired ServiceNow, Workday, and Microsoft Graph APIs, enabling seamless enterprise data flow across multiple internal IT platforms.
- Configured **PostgreSQL** vector search and **Redis** caching for sub-200ms latency, cutting release cycles by 40% with **Jenkins Docker CI/CD**.

Alibaba Group — Guangzhou, China — Jun. 2025 – Aug. 2025
Software Development Intern — ***React, TypeScript, GitLab CI/CD, Jest***

- Refactored a legacy mail-dispatch module with **React, TypeScript**, and Ant Design Form, removing DOM scripts and cutting bug reports by 40%.
- Debugged production email failures with unit tests and guarded submission flows, restoring 100% send-success SLA before release deadlines.
- Accelerated front-end rendering via memoization, key-based re-mounting, and dynamic imports, cutting first-paint time by 30%.
- Established **GitLab CI/CD** with Jest/RTL suites at 90% coverage, shrinking review turnaround from 2 days to 24 hours.

Guangzhou Risong Intelligent Technology — Guangzhou, China — Jun. 2024 – Aug. 2024
Computer Vision & AI Intern — ***Python, PyTorch, OpenCV, Embedded GPU***

- Engineered scalable deep learning image-processing pipelines with **Python and OpenCV** for industrial robot vision in automotive welding lines.
- Optimized CNN models with **PyTorch** for object localization and surface quality inspection, improving detection precision by 12% over baseline.
- Packaged inference results into robot control software for automated welding parameter adjustment based on real-time visual feedback.
- Deployed optimized models to embedded GPU devices, achieving sub-200ms inference latency per frame in production environments.

PROJECTS

Online Order – Spring Boot Food Ordering System — ***Spring Boot, React, PostgreSQL, AWS, Spring Security, CI/CD***

- Designed CRUD **REST APIs** with **Spring Boot** Controllers covering registration, menu search, ordering, payment, and checkout workflows.
- Configured **Spring Data JDBC** with **PostgreSQL** on **AWS RDS**, structuring controller/service/repository layers with dependency injection.
- Secured the platform via **Spring Security** and containerized the full-stack service to **AWS ECR** with automated **CI/CD** workflows.

Social AI – AI-Based Social Network — ***React, Go, JWT, Elasticsearch, GCP, DALL-E 3***

- Built a **React**-based social network with DALL-E 3 integration, enabling AI-assisted post creation and image generation for concurrent users.
- Implemented **JWT**-based authentication with React Router v4 and server-side session management for secure user access and authorization.
- Deployed a **Go** backend on Google App Engine, integrating **Elasticsearch** on GCE for fast real-time full-text social media search.

PUBLICATIONS

Attribute-Based Interpretable Incremental Learning for Blood Cell Image Recognition — MICCAI 2025 (Under Review) **[PDF]**